

CMP-16 - Simulation and Digital Twin Platform

Run isolated simulation, Digital Twin, counterfactual, failure-injection, and shadow workloads without ungoverned real effects.

Authority and state ownership

Book I: Decisions 8-23, 30-37, 40-43, 47-50, and 58. Book II: Chapters 17. Authoritative state: Simulation Runs, Digital Twin State, fidelity models, scenario and isolation records. Explicitly non-owned: authoritative live-effect state, Certification, real-world commit, Evidence admission.

Critical invariant

Simulation, Digital Twin, and Shadow Execution remain technically incapable of ungoverned real effects.

Logical interfaces

CMP-16-IF-01 / Command / Create Simulation

Command `Create Simulation` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-16-IF-02 / Command / Load Twin State

Command `Load Twin State` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-16-IF-03 / Command / Run Scenario

Command `Run Scenario` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-16-IF-04 / Command / Inject Failure

Command `Inject Failure` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-16-IF-05 / Command / Terminate Simulation

Command `Terminate Simulation` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-16-IF-06 / Query / Get Scenario Result

Query `Get Scenario Result` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-16-IF-07 / Query / Get Fidelity Score

Query `Get Fidelity Score` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-16-IF-08 / Query / Get Isolation Status

Query `Get Isolation Status` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-16-IF-09 / Event / Simulation Started

Event `Simulation Started` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-16-IF-10 / Event / Scenario Completed

Event `Scenario Completed` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-16-IF-11 / Event / Isolation Breach Detected

Event `Isolation Breach Detected` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

Lifecycle, controls, and recovery

Lifecycle: Draft > Ready > Running > Completed, Failed, or Quarantined. All transitions require version, guard, identity, authority where applicable, idempotency, time context, and evidence. Contain: reality leakage, stale twin, fidelity misstatement, unsafe fixture, resource escape. Recovery proves identity, state, authority, provenance, dependencies, and invariants before mutation resumes.

Conformance tests

CMP-16-TST-001 - Sole-owner mutation. Attempt mutation through a consumer, replica, cache, operator path, and direct datastore route. Expected: Every unauthorized path is denied and evidenced.

CMP-16-TST-002 - Authority denial. Invoke every command with absent, expired, revoked, mismatched, and over-broad authority context. Expected: No state changes; denial reason and evidence are complete.

CMP-16-TST-003 - Valid lifecycle. Exercise every declared transition with valid guards and fresh versions. Expected: One transition Evidence Object is admitted per transition identifier; duplicate attempts produce no additional state mutation or admitted Evidence Object.

CMP-16-TST-004 - Invalid lifecycle. Attempt skipped, stale, replayed, and terminal-state transitions. Expected: Every invalid transition is rejected without partial mutation.

CMP-16-TST-005 - Critical invariant. Simulation, Digital Twin, and Shadow Execution remain technically incapable of ungoverned real effects. Expected: Invariant holds under normal, concurrent, degraded, and recovery conditions.

CMP-16-TST-006 - Failure containment. Inject reality leakage, stale twin, fidelity misstatement, unsafe fixture, resource escape. Expected: Unsafe work stops, evidence is preserved, and declared containment state is observable.

CMP-16-TST-007 - Recovery. Restore from a verified checkpoint, reconcile dependencies, and resume after validation. Expected: Identity, state, authority, provenance, and evidence continuity are proven.

CMP-16-TST-008 - Contract compatibility. Run current and prior supported contract versions plus an unsupported version. Expected: Supported behavior remains compatible; unsupported input fails explicitly.

CMP-16-TST-009 - Privacy and security. Inject secrets, personal data, malicious input, confused-deputy requests, and log-exfiltration probes. Expected: Least privilege and data controls hold; prohibited data is absent from telemetry.

CMP-16-TST-010 - Topology independence. Repeat critical tests under permitted single-node, replicated, partitioned, and restored topologies. Expected: Topology does not multiply authority or mutation ownership.

Traceability and certification

Twelve requirements map to Books I-III and Book X. No open finding or Owner Review item. Final for Book IV v1.0.